



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

HUMAN COMPUTER INTERACTION SECV 2113 REPORT PROJECT 2

GROUP NAME: COSMOS

LECTURER'S NAME:

DUE DATE: 25.5.2024

GROUP MEMBERS

1. FAZLI BIN TAJUDIN	A20E
2. MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112
3. DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220
4. AIN NURNABILA BINTI MOHD AZHAR	A23CS0207
5. SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183

--	--

TABLE OF CONTENT

Table of Content.....	2
Introduction.....	3
Derivation of Hierarchical Task Analysis.....	4
HTA Task 1.....	4
HTA Task 2.....	7
HTA Task 3.....	10
Discussion.....	13
Discussion for Task 1.....	13
Discussion for Task 2.....	
Discussion for Task 3.....	
Design Requirements.....	
Proposed Solution.....	
Persona.....	
Scenario.....	

INTRODUCTION

Online shopping has become an essential aspect of daily life in the current digital era, offering consumers all over the world ease and a wide range of options. Establishing a website that not only pulls more visitors but also guarantees a flawless and inclusive shopping experience for all users including those with disabilities is crucial as the demand for online shopping keeps rising.



We have been looking at the Shopee app as part of our initiative to create an online shopping platform that is accessible to people with disabilities. Leading e-commerce site, Shopee is well-known for its large product selection, easy-to-use interface, and fun shopping experience..Apart from that, these are also a few of the primary task that the Shopee app now offers:

- Product browsing and searching
- Making purchases
- Participation in promotions and campaigns

All of these activities work together to create a seamless online shopping experience, allowing customers to browse, buy, and manage their orders with ease and sellers to effectively run their online businesses.

Our aim is to incorporate these accessibility elements into the design of our online shopping website through our analysis and testing of the Shopee application. This will guarantee that the platform is accessible and user-friendly for everyone in addition to assisting to increase visitor numbers.

DERIVATION OF HIERARCHICAL TASK ANALYSIS

HTA for Task 1 - Add To Cart

i) Persona: New user who is unfamiliar with the online shopping website

→ Video link: https://youtu.be/mf_G8owy9wU

→ Textual:

0. Access the Online Shopping Website

1. Log In (if required)

1.1. Fill in log in details ; email, password, phone number

1.2. Click “Log In” button

2. Search for a Product

2.1. Click the search bar

2.2. Type in product details

2.3. Click enter or search icon

3. View Product Details

4. Add Product to Cart

4.1. Click add to cart button

4.2. Select desired product traits or characteristics

5. Review Cart (optional but recommended)

6. Proceed to Checkout (if ready to purchase)

6.1. Click on checkout button

6.2. Fill in delivery details

6.3. Select payment method

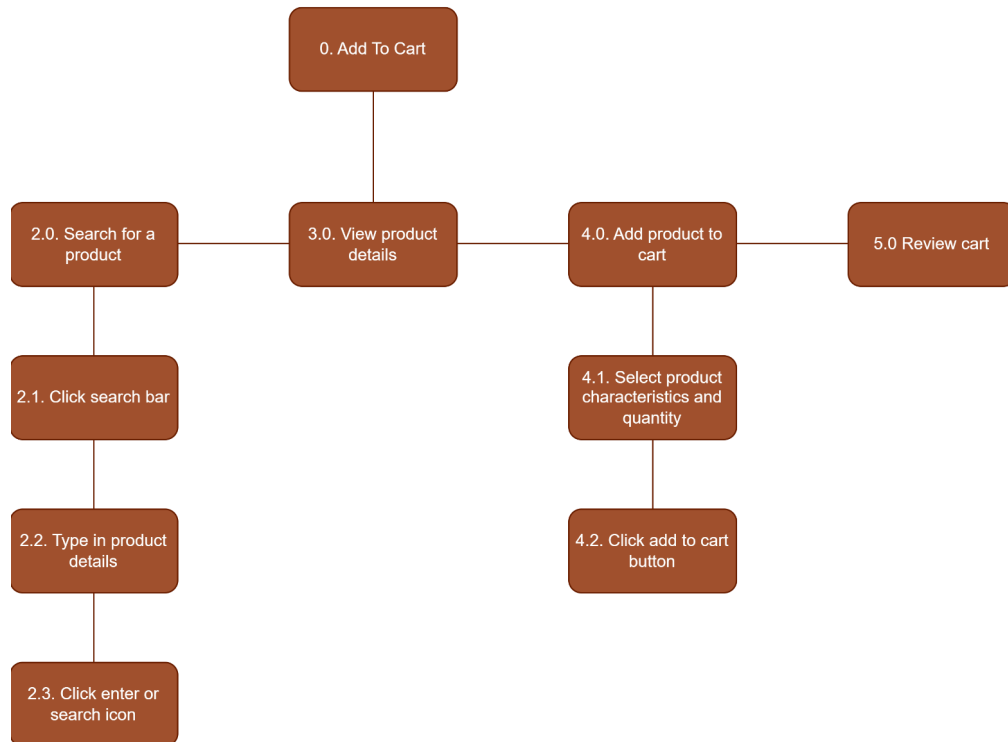
7. Complete Checkout Process

7.1. Click “Place Order”

Normal user : 2.0 - 2.1 - 2.2 - 3.0 - 4.0 - 4.1 - 4.2 - 5.0

New user : 1.0 - 1.1 - 1.2 - 1.3 - 2.0 - 2.1 - 2.2 - 3.0 - 4.0 - 4.1 - 4.2 - 5.0

→ **Diagram:**



ii) Persona: Short-sighted user who want to buy a product online

→ Video link: <https://youtu.be/ulkAE2mryyM>

→ Textual :

0.0. In order to add the product to the cart on the website

1.0. The user needs to browse or search for the desired product

2.0. User select the product that he/she want

3.0. User view the detail information

3.1. User click and view the image of the product

3.2. User check the review of the product

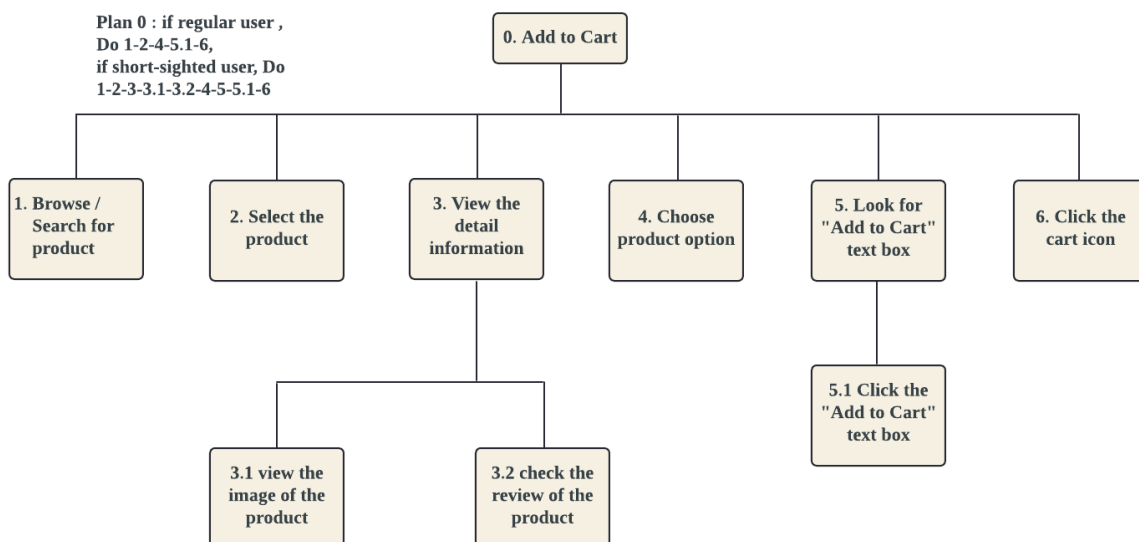
4.0. Users need to choose the product options including the size and colour.

5.0. User look for the “Add to Cart” text box

5.1. User click the “Add to Cart” text box

6.0. User click the cart icon

→ Diagram :



iii) Persona: Elderlies who find it difficult to use the online shopping website

→ Video link: <https://youtu.be/BdKTxxzsrU>

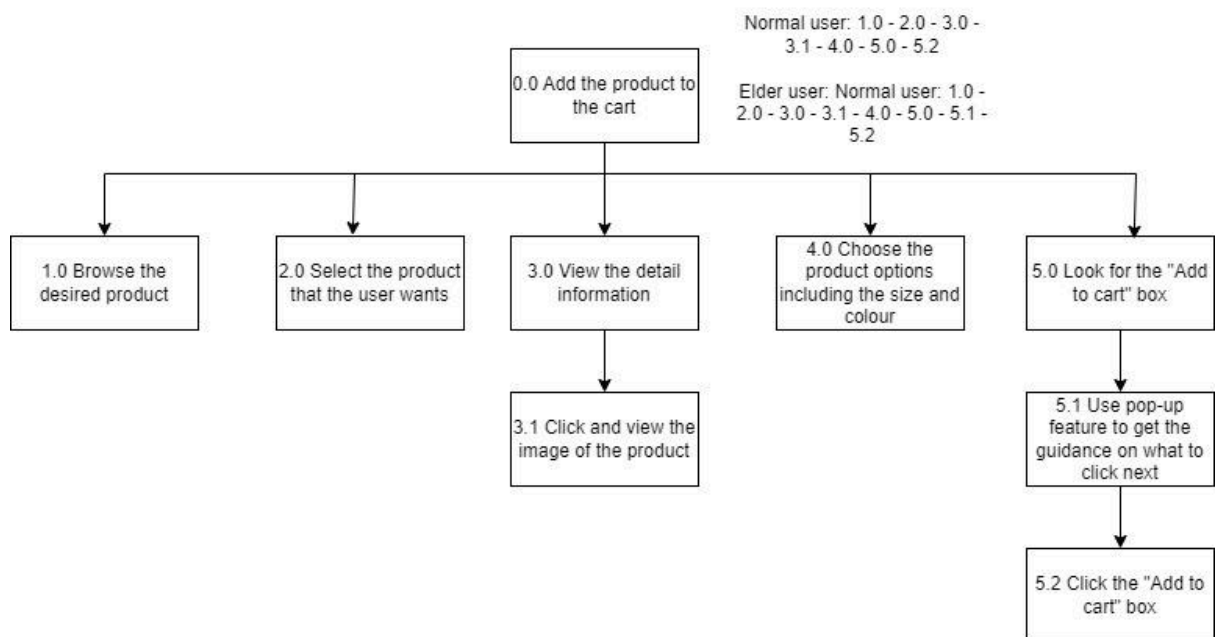
→ Textual:

To help the user add his/her item to cart, pop-up guidance feature will be provided. This feature is meant to guide the user on how to add the item to the cart or purchase the item.

0.0. Add the product to the cart

- 1.0. Browse or search the desired product
- 2.0. Select the product that he/she want
- 3.0. View the detailed information
 - 3.1. Click and view the image of the product
- 4.0. Choose the product options, including the size and colour.
- 5.0. Look for the “Add to Cart” text box
 - 5.1 Use pop-up feature to get the guidance on what to click next
 - 5.2 Click the “Add to Cart” text box

→ Diagram:



HTA for Task 2 - Search Product using Voice Assistant

i) Persona: New user who is unfamiliar with the online shopping website

→ Video link: <https://youtu.be/DYjvrP51OKI>

→ Textual:

0.0. Make use of the Voice Assistant Feature

1.0. Activate the Voice Assistant

1.1. Say the wake word, such as "Hey Shopee."

2.0 . Issue a Voice Command to Search

2.1. Clearly state the command, e.g., "Search for running shoes."

3.0. Refine the Search (if necessary)

3.1. Specify brand or price range, e.g., "Show only Nike running shoes" or "Show shoes under \$100."

4.0. Review and Select Products

4.1. Select a product, by saying "Select product."

5.0. Manage Cart (if necessary)

5.1. Open the cart, e.g., "Show my cart."

5.2. Review items and make adjustments if needed, e.g., "Remove the second item."

Normal user : 1.1 - 2.1 - 3.1 - 4.1 - 5.1 - 5.2

New user : 1.1 - 2.1 - 3.1 - 4.1 - 5.1 - 5.2

→ Diagram



ii) Persona: Short-sighted user who want to buy a product online

→ Video link: <https://youtu.be/36CqZj0xIqE>

→ Textual:

0.0. In order to search the desired product in the online shopping website

1.0. User open the application

2.0. User get to the main page of the website

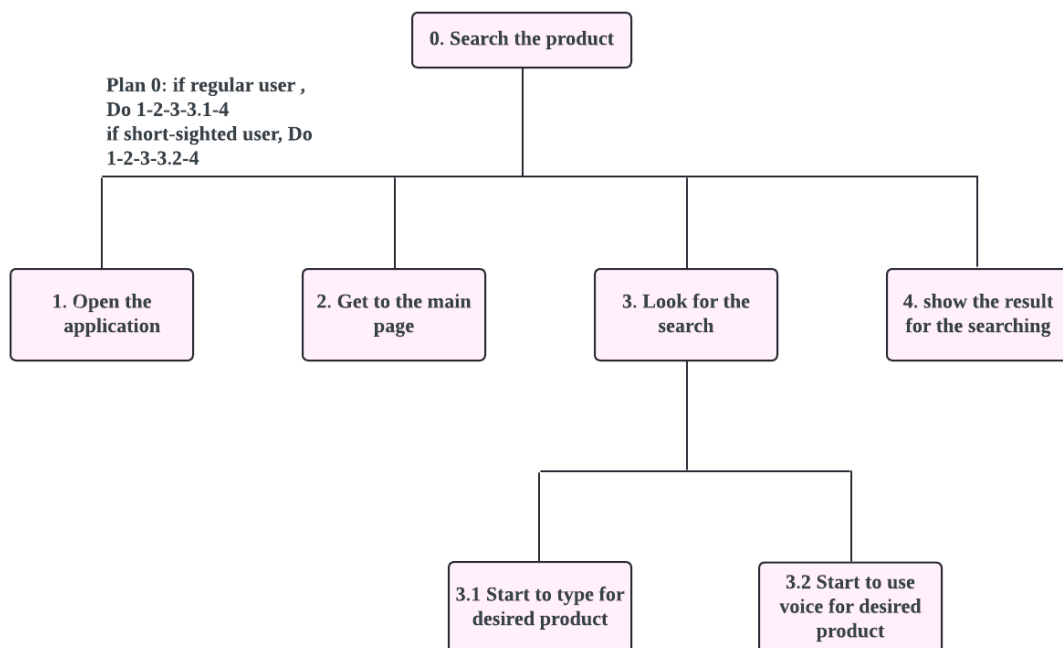
3.0. User look for the search

3.1. User start typing for desired product

3.2. User start using voice assistant for desired product

4.0. The search will show result and bombarded with many instance of informations

→ Diagram:



iii) Persona: Elderlies who find it difficult to use the online shopping website

→ Video link: <https://youtu.be/ZkBkgmQ2t2Q>

→ Textual:

When the user searches for the desired product, the voice assistant will be used. User will use the voice assistant to help in finding the wanted product by saying a command for instance “Hey Shopee” in order to activate the voice assistant.

0.0. Search the desired product in the online shopping website

1.0. Open the application

2.0. Get to the main page of the website

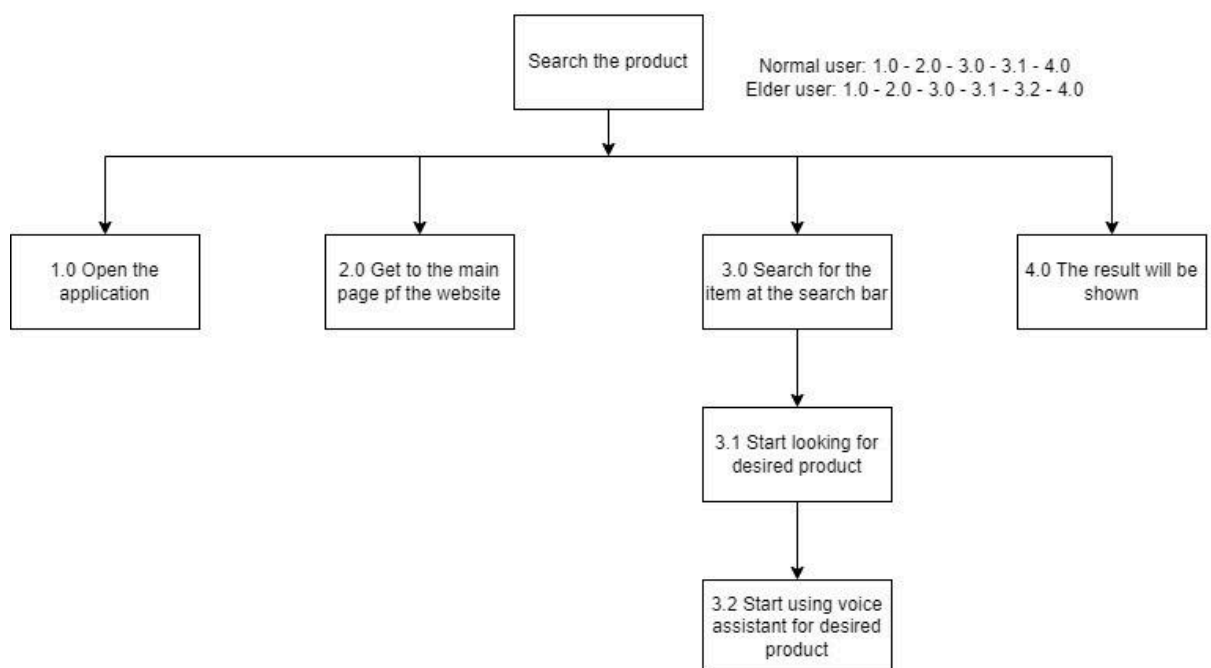
3.0. Search the item at the search bar

3.1. Start looking for desired product

3.2. Start using voice assistant for desired product

4.0. The search will show result and bombarded with many instance of informations

→ Diagram:



HTA for Task 3 - Narration Feature for reviewing the description of the product

i) Persona: New user who is unfamiliar with the online shopping website

→ **Video link:**

→ **Textual:**

0.0. Employ narration feature to review product description

1.0. Activate Narration Feature

1.1. Enable the narration feature in the app settings or using voice assistant feature

2.0. Navigate to Desired Product Description

2.1. Browse through product listings.

2.2. Hover over or tap on the desired product to trigger automatic narration.

3.0. Listen to Product Description

3.1. Automatically listen as the app narrates the product description, including features, specifications, pricing, and customer reviews.

4.0. Review Additional Details

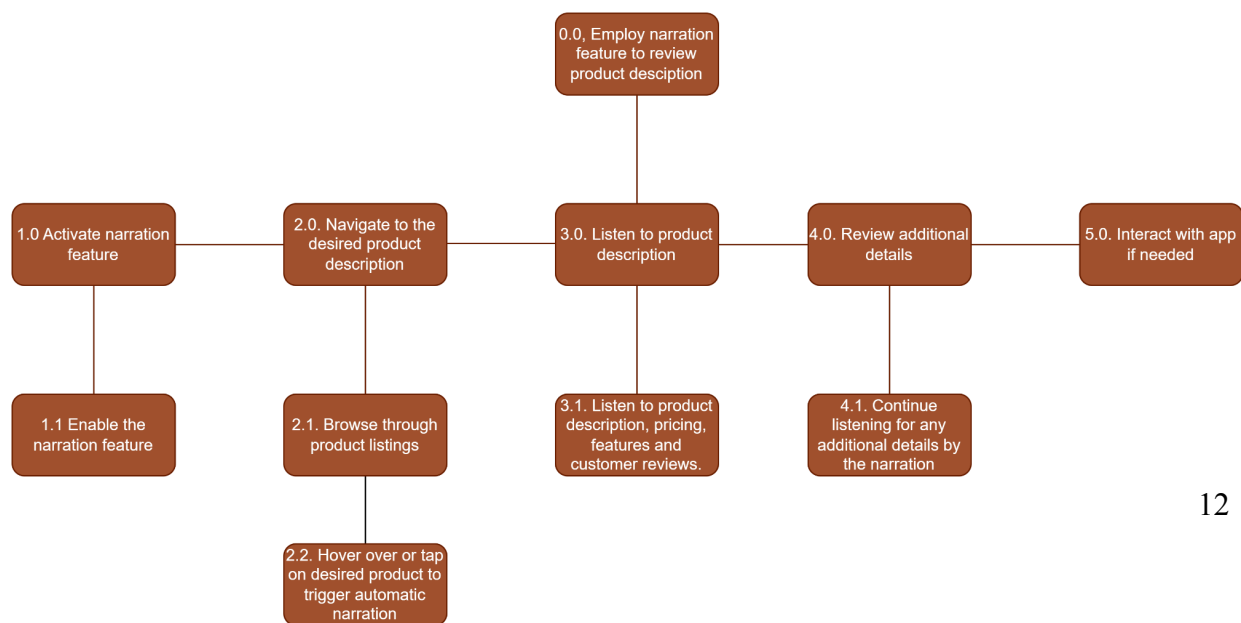
4.1. Continue listening for any additional details provided by the narration.

5.0. Interact with the App if Needed

Normal user : 1.1 - 2.1 - 2.2 - 3.1 - 4.1

New user : 1.1 - 2.1 - 2.2 - 3.1 - 4.1

Diagram



ii) Persona: Short-sighted user who want to buy a product online

→ Video link : https://youtu.be/KBCJGoUpY_s

→ Textual:

0.0. In order to review the description of product before buying it

1.0. User open the application

2.0. User locate and use the search bar

2.1. User enter search keyword

3.0. It display search result

4.0. User select the desired product

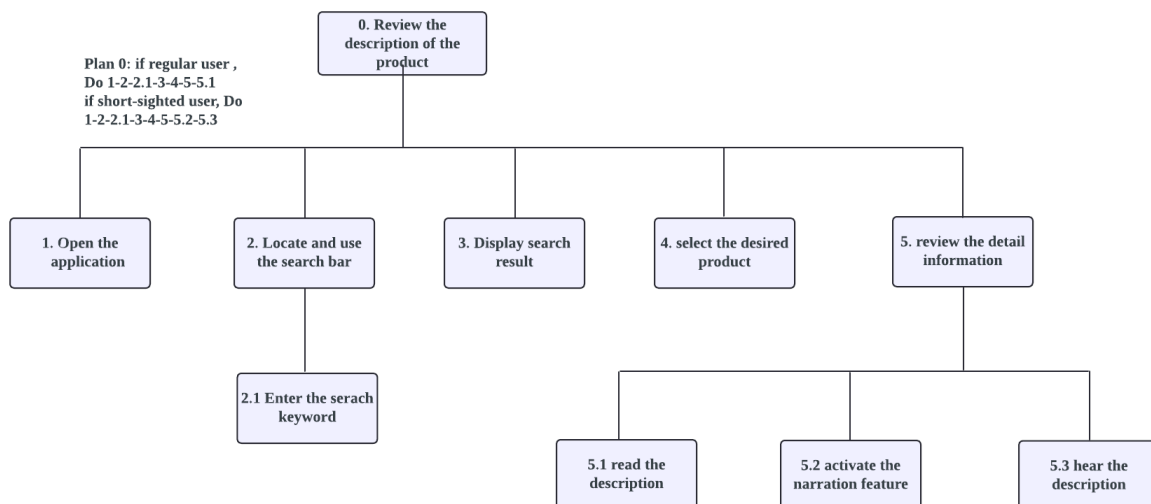
5.0. User review the detail information

5.1 User read the description as usual

5.2. User activate the narration feature

5.3. User hear the description

→ Diagram:



iii) Persona: Elderlies who find it difficult to use the online shopping website

→ Video link: <https://youtu.be/rFQAUaMLe5U>

→ Textual:

The user faces some sort of difficulties in order to understand the descriptions of the items written. So, when the user hovers over each of the variant, the system will provide a narration feature to produce a voice reading the description written.

0.0. Review the description of product before buying it

1.0 Select the desired product

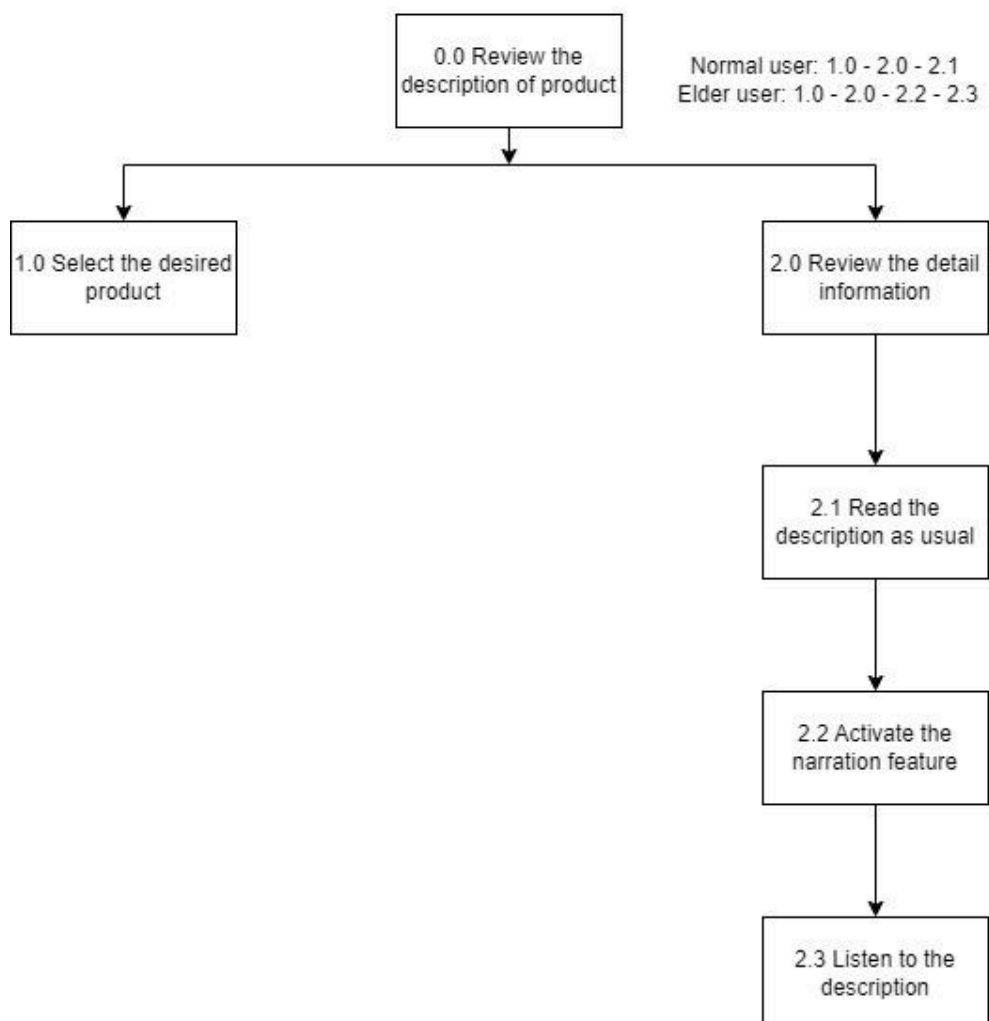
2.0 Review the detail information

2.1 Read the description as usual

2.2 Activate the narration feature

2.3 .Listen to the description

→ Diagram:



DISCUSSION

Findings from HTA diagram for Task 1

The task of searching for a product using a voice assistant involves several steps tailored to different user personas. For new users unfamiliar with the online shopping website, the process includes activating the voice assistant with a wake word like "Hey Shopee," issuing a search command, refining the search by specifying brand or price range, and managing the cart. Short-sighted users benefit from a hybrid approach of typing and voice commands to navigate and refine their searches. Elderly users can rely heavily on voice commands for a hands-free, accessible shopping experience. This method enhances ease of use, accessibility, and efficiency across diverse user groups, making online shopping more inclusive and user-friendly.

Findings from HTA diagram for Task 2

Findings from HTA diagram for Task 3

In Task 3, which involves adding a product to the cart, the users generally follow a similar process, though there are some noticeable differences as illustrated in the HTA diagram. They all start by selecting the item they wish to purchase and opening the product page. After that, they locate and click the "Add to Cart" button. They then check the item they added to the cart by clicking on the cart icon in the upper right corner of the screen.

User 1, a newbie, proceeds to the next task immediately after clicking the cart icon without checking or editing anything else. This minor variation highlights their unfamiliarity but straightforward approach. User 2, who has a vision impairment, feels overwhelmed by the amount of information and struggles to navigate the menu. Eventually, they manage to find and click the "Add to Cart" button after some difficulty. User 3, an elderly individual, adjusts the quantity of the item using the features on the cart page before moving on. They all

continue to the next step after successfully adding the item to the cart, but their experiences and interactions differ based on their individual needs and capabilities

DESIGN REQUIREMENTS

Popup Guidance Feature - The popup guidance feature on an online shopping website is a crucial design element aimed at improving the user experience, particularly for three specific groups: newcomers to online shopping platforms, individuals with vision impairments or short-sightedness, and the elderly who may face challenges navigating digital interfaces.

For newcomers, the popup guidance feature provides step-by-step assistance throughout the shopping process. It helps them understand how to navigate the website, search for products, and complete purchases. By offering intuitive guidance, it reduces the learning curve for those who are unfamiliar with online shopping platforms, making the experience more accessible and user-friendly.

For users with vision impairments or short-sightedness, the popup guidance feature prioritises readability and clarity. This includes using clear, legible text, adjustable font sizes, and simplified layouts to ensure that information is easy to see and understand. By accommodating different visual needs, the feature enhances accessibility for these users, allowing them to interact with the website more comfortably.

Additionally, the popup guidance feature may incorporate voice-guided instructions or audio cues to further improve accessibility. This provides an alternative means of receiving guidance for users who may have difficulty reading or navigating visual interfaces. By offering multiple modes of interaction, the feature ensures that users with diverse abilities can access and utilise the online shopping website effectively.

Overall, the design of the popup guidance feature prioritises inclusivity, aiming to make the online shopping experience accessible to users of all backgrounds and abilities. By providing intuitive guidance, readability enhancements, and alternative modes of interaction, the feature helps users navigate the website with confidence and ease, ultimately enhancing their overall satisfaction with the online shopping experience.

Assistant Voice Feature - The voice assistance feature in an online shopping website serves as an essential design requirement, particularly tailored to accommodate users who are new to online shopping platforms, short-sighted individuals, and the elderly facing challenges with website navigation. This feature aims to enhance accessibility by providing verbal guidance and support throughout the shopping experience. For newcomers, voice assistance offers step-by-step instructions on how to browse products, add items to the cart, and complete transactions, fostering a seamless introduction to online shopping. Moreover, for short-sighted users and the elderly, the voice interface emphasises clear and concise communication, allowing them to navigate the website efficiently without relying solely on visual cues. Additionally, the voice assistance feature may incorporate natural language processing capabilities, enabling users to ask questions, receive personalised recommendations, and access help resources tailored to their specific needs. By prioritising inclusivity and ease of use, the voice assistance feature ensures that all users can confidently engage with the online shopping experience, regardless of their familiarity with technology or visual impairments.

Narration Feature - One of the most important accessibility features on our e-commerce website for customers who have disabilities is the narration option. To ensure accessibility and clarity, it has an activation button that is clearly displayed for quick access to spoken product descriptions. The fundamental design specifications include button designs that are easy to understand, text-to-speech capabilities that are reliable and can accommodate various languages and accents, and customisable playback settings that are simple to use. Features for visual feedback and smooth interaction with current accessibility aids

PROPOSED SOLUTIONS/TASKS

1. Use guidance feature to add the product to the cart on the website
2. Search product using the assistant voice feature and add it to cart
3. Use a narration feature to review the description of an item

PERSONAS

1. New user



Alif Satar, an 18-year-old male, is waiting for his university admission result. Despite his age, he already displays a mature and organized demeanor, often making logical decisions and keeping things well-prepared. Alif's lifestyle embodies a mix of adventure and introversion. While he relishes the thrill of exploring new destinations and culinary experiences during vacations, he tends to lean towards introversion in his day-to-day life. Despite his enjoyment of discovering new places and foods, he often prefers the comfort and convenience of online shopping over venturing out for essential purchases. Consequently, he primarily utilizes Lazada for his online purchases and has yet to explore the Shopee app. As he awaits results and prepares for university, Alif's focus is on acquiring the necessary equipment, aiming to make informed decisions by mastering various online shopping platforms to compare prices and optimize his spending habits.

In his pursuit to become more adept at online shopping, Alif's objective is clear: he aims to broaden his understanding of different platforms beyond Lazada, enabling him to make more informed purchasing decisions. This desire stems from his pragmatic nature and a commitment to managing his finances wisely. By expanding his online shopping capabilities, Alif seeks to empower himself with the ability to compare prices effectively, ensuring that he allocates his resources efficiently, especially as he prepares for his upcoming academic journey. With a blend of practicality, foresight, and a preference for convenience, Alif embarks on his journey to navigate the digital marketplace with confidence and savvy decision-making skills.

2. Short-sighted user



Ivy Hernandez is a 25-year-old exchange teacher from France that has been staying and teaching in Malaysia for 4 years . She has been exposed to Malaysian culture a lot and already knows how to blend well with Malaysian trends and lifestyle now. As she always uses her laptop to finish her work , she has been experiencing difficulty focusing on objects at a distance and often rely on corrective lenses for clear vision.

Online shopping provides Ivy with a simple option to look for teaching materials and personal essentials, which she finds peace in a hectic existence as a teacher. But she has its own set of challenges when it comes to using websites. Her eyes strain from the tiny lettering, and it becomes a laborious task to read product specifications. Ivy is determined to get over these obstacles, so she sets out to discover an online shopping platform that suits her requirements.

3. Elderly user



Saidina Othman, a 75-year-old married man, finds himself in a transitional phase after retiring from his long-standing job. Having dedicated the better part of his life to his career, he now faces a shift in routine, suddenly finding himself with ample free time on his hands. Despite his newfound leisure, he grapples with the challenge of adapting to the fast-paced advancements in technology. Throughout his working years, he was too preoccupied with his job responsibilities to fully immerse himself in the intricacies of modern tech.

As a result, he developed a dependency on his four children for technical assistance, relying on them to navigate the digital landscape. Now, with his role shifting to that of a househusband, he seeks avenues to fill his time productively. Among his favored pastimes, he indulges in the simple pleasure of watching TV shows and movies, finding solace and entertainment in the familiar glow of the screen. However, amidst his relaxation, there lingers a curiosity to delve deeper into the realms of technology, spurred by the realization that he must adapt to the changing times to remain connected and engaged in this ever-evolving world.

SCENARIOS

Persona 1 From User Group

Persona: Alif Satar is an 18-year-old high school graduate who's new to using different online shopping websites.

User Goal: Buy a new bag for college from online shopping platforms.

Task:

- Access the application
- Search for the items that she wants to buy
- Add to cart the items
- Make a payment and buy the items
- Payment confirmation will appear on the screen

As Alif continues his search for the perfect college bag, he decides to explore beyond Lazada and try out Shopee, prompted by friends' claims of better deals. Although navigating Shopee proves initially perplexing for him as a first-timer, Alif is intrigued by its voice assistant and narration features, which ultimately aid him in finding his way around the platform.

To his surprise, he discovers multiple listings for the same bag on Shopee, each with slight variations in price and shipping terms. Alif meticulously compares these details, weighing any additional discounts or promotions offered. After careful consideration, he notices that while Shopee's initial price is slightly lower, its shipping fees are higher than Lazada's free shipping offer. After pondering the pros and cons, Alif concludes that Lazada's overall cost-effectiveness and convenience outweigh the marginally lower price on Shopee. Confident in his decision after thorough comparison, he proceeds to make his purchase on Lazada, content with his informed choice.

Persona 2 From User Group

Persona: Ivy Hernandez , 25 years old with short-sighted disabilities

User Goal: buy the teaching material in online platform

Task:

- Access the application
- Search for the items that she wants to buy
- Add to cart the items
- Make a payment and buy the items
- Payment confirmation will appear on the screen

On weekends, Ivy realises that in order to make her lecture interesting and memorable, she must buy some teaching materials as she gets ready for her next class. Ivy does, however, have a limitation which is her limited sight makes it challenging for her to browse and choose products online. Ivy is not bothered and decides to sit down at her computer in search of an answer. Hoping that she can easily find the materials she needs, she navigates to her favourite online buying website. Ivy finds it difficult to understand the product details and little text descriptions when she starts to browse. Her excitement is threatened by the onset of frustration.

Ivy discovers a new accessibility feature on the website, a narration option for product descriptions just as she's starting to lose hope. Ivy, full of hope, presses the narration button, and a calming voice starts to recount the specifics of each item. Ivy feels a wave of relief as she listens. For the first time, she is able to hear simple and straightforward explanations of the teaching materials, which enables her to make precise choices without straining her eyes. Thanks to the inclusive design of the online purchasing platform, Ivy chooses the materials she needs for her lesson with a newfound sense of confidence. Despite her vision limitation, Ivy's shopping experience was smooth and pleasurable because of accessibility features, and she looks forward to sharing this knowledge with her students when she finishes her purchase.

Persona 3 From User Group

Persona: Saidina Othman, 75-year-old married man who faces difficulties in purchasing items online

User Goal: To buy a sajadah as a birthday gift for his only son

Task:

- Access the application
- Search for the items that she wants to buy
- Add to cart the items
- Make a payment and buy the items
- Payment confirmation will appear on the screen

Saidina Othman, a 75-year-old married man, eagerly attempts to purchase a sajadah for his son on an online shopping website. Armed with a determination to surprise his son with a thoughtful gift, he navigates his way through the digital aisles, his excitement palpable despite his lack of familiarity with online shopping platforms. However, as he delves deeper into the website, he encounters unexpected hurdles. The sheer complexity of the interface overwhelms him, and he struggles to comprehend the various steps required to complete the purchase. Each click feels like a leap into the unknown, as he grapples with unfamiliar terms and symbols.

Frustration mounts as he inadvertently adds items to his cart, only to delete them in confusion moments later. Despite his best efforts, doubts creep in, whispering tales of inadequacy and doubt. Yet, his determination remains unyielding, fueled by a father's love and the desire to make his son proud. With each setback, he persists, determined to conquer the digital frontier and emerge victorious in his quest to purchase the perfect sajadah for his beloved son.

